

# Mathematics for Data Scientists

Ahmad TAY

EFREI-Paris

Autumn 2024

- 1 Linear Algebra
  - Introduction
  - Linear Systems
  - Basic matrix operations (+, -, \*)
  - Determinants and inverses
  - Vector Spaces and Linear Transformations
- 2 Analytic Geometry
  - Normed Vector Spaces
  - Unit vectors and orthonormal basis
- 3 Matrix Decomposition
  - Diagonalisation
  - Analyse en Composantes Principales
- 4 Calculus and Optimization
  - Differentiation
  - Gradient descent
- 5 Probability Theory
- 6 Statistical Modeling
- 7 Statistical Inference

# What is Vector Calculus?

Compute gradients of functions, which is often essential to facilitate learning in machine learning models.

- **Vector Calculus** is a branch of mathematics focused on differentiation and integration of vector fields.
- It extends the concepts of calculus from functions of one variable to functions of multiple variables.
- differential calculus is about finding the rates of change of related quantities.
- Key topics include:
  - Vector fields
  - Gradient, Divergence, and Curl
  - Line, Surface, and Volume Integrals

## Example: Gradient of a Scalar Field

- Consider the scalar field  $f(x, y, z) = x^2 + y^2 + z^2$ .
- The **gradient** of  $f$ , denoted by  $\nabla f$ , is a vector field that points in the direction of the greatest rate of increase of  $f$ .

### Calculation

$$\nabla f = \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$\nabla f = \left( \frac{\partial}{\partial x}(x^2 + y^2 + z^2), \frac{\partial}{\partial y}(x^2 + y^2 + z^2), \frac{\partial}{\partial z}(x^2 + y^2 + z^2) \right)$$

$$\nabla f = (2x, 2y, 2z)$$

- The gradient vector field is  $\nabla f = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$ .

# Derivative

For  $h > 0$ , the **derivative** of a function  $f$  at a point  $a$  is defined as:

$$\frac{df}{da} = f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

We want to compute the derivative of  $f(x) = x^n$ ,  $n \in \mathbb{N}$ . We may already know that the answer will be  $nx^{n-1}$ , but we want to derive this result using the definition of the derivative as the limit of the difference quotient.

Using the definition of the derivative in (5.4), we obtain

$$\begin{aligned} \frac{df}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sum_{i=0}^n \binom{n}{i} x^{n-i} h^i - x^n}{h} \end{aligned} \tag{5.5c}$$

We see that  $x^n = \binom{n}{0}x^n$  and, by starting the sum at 1, the  $x^n$  terms cancel, and we obtain

$$\begin{aligned}
 \frac{df}{dx} &= \lim_{h \rightarrow 0} \frac{\sum_{i=1}^n \binom{n}{i} x^{n-i} h^i}{h} \\
 &= \lim_{h \rightarrow 0} \sum_{i=1}^n \binom{n}{i} x^{n-i} h^{i-1} \\
 &= \lim_{h \rightarrow 0} \binom{n}{1} x^{n-1} + \underbrace{\sum_{i=2}^n \binom{n}{i} x^{n-i} h^{i-1}}_{\rightarrow 0 \text{ as } h \rightarrow 0} \\
 &= \frac{n!}{1!(n-1)!} x^{n-1} \\
 &= nx^{n-1}
 \end{aligned}$$

# Differentiation rules: Sum

Differentiation is a linear operator, so we can calculate the derivatives term by term. We can then define the *sum rule* states that for two functions  $f(x)$  and  $g(x)$ :

$$\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x) = f'(x) + g'(x)$$

## Example

Let's consider the functions  $f(x) = x^2$  and  $g(x) = 3x$ . According to the sum rule for differentiation:

$$\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}(x^2 + 3x) = \frac{d}{dx}(x^2) + \frac{d}{dx}(3x) = 2x + 3$$

# Product

The product rule for differentiation states that for two functions  $f(x)$  and  $g(x)$ :

$$\begin{aligned}\frac{d}{dx}(f(x) \cdot g(x)) &= f(x) \cdot \frac{d}{dx}g(x) + g(x) \cdot \frac{d}{dx}f(x) \\ (f(x)g(x))' &= f(x)g'(x) + f'(x)g(x)\end{aligned}$$

## Example

Let's consider the functions  $f(x) = x^2$  and  $g(x) = 3x$ . Applying the product rule:

$$\frac{d}{dx}(x^2 \cdot 3x) = x^2 \cdot \frac{d}{dx}(3x) + 3x \cdot \frac{d}{dx}(x^2)$$

Differentiating each term:

$$= x^2 \cdot 3 + 3x \cdot 2x = 3x^2 + 6x^2 = 9x^2$$

# Quotient rule

The quotient rule for differentiation states that for two functions  $f(x)$  and  $g(x)$ , where  $g(x) \neq 0$ :

$$\begin{aligned}\frac{d}{dx} \left( \frac{f(x)}{g(x)} \right) &= \frac{g(x) \cdot \frac{d}{dx} f(x) - f(x) \cdot \frac{d}{dx} g(x)}{[g(x)]^2} \\ &= \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}\end{aligned}$$

## Example

$$\frac{d}{dx} \left( \frac{x^2}{3x} \right) = \frac{3x \cdot \frac{d}{dx} (x^2) - x^2 \cdot \frac{d}{dx} (3x)}{(3x)^2}$$

Differentiating each term:

$$= \frac{3x \cdot 2x - x^2 \cdot 3}{9x^2} = \frac{6x^2 - 3x^2}{9x^2} = \frac{3x^2}{9x^2} = \frac{1}{3}$$

# Chain Rule

Two functions are composed when the output of one is used as the input to another.

Let  $f$  and  $g$  be two functions.

The chain rule (applies to function compositions) for differentiation states that the derivative of  $g \circ f$  and  $f \circ g$  with respect to  $x$  is given by:

$$(g(f(x)))' = \frac{d}{dx}(g \circ f(x)) = g'(f(x)) \cdot f'(x)$$

and

$$(f \circ g)'(x) = \frac{d}{dx}(f \circ g(x)) = f'(g(x)) \cdot g'(x)$$

## Remark

The chain rule is of fundamental importance in the training of neural networks.

### Example: $g \circ f$

Let  $f(x) = x^2$  and  $g(x) = 3x + 1$ .

First, differentiate  $g(x)$  with respect to  $x$ :

$$g'(x) = 3$$

Next, differentiate  $f(x) = x^2$  with respect to  $x$ :

$$f'(x) = 2x$$

Now, apply the chain rule:

$$\frac{d}{dx} (g(f(x))) = g'(f(x)) \cdot f'(x)$$

Substituting the values:

$$\frac{d}{dx} (3(x^2) + 1) = 3 \cdot 2x = 6x$$

## Example $f \circ g$

First, differentiate  $f(x) = x^2$  with respect to  $x$ :

$$f'(x) = 2x = 2(3x + 1)$$

Next, differentiate  $g(x) = 3x + 1$  with respect to  $x$ :

$$g'(x) = 3$$

Therefore, by the chain rule:

$$\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x) = 2(3x + 1) \cdot 3 = 6(3x + 1)$$

# Partial Derivatives

Consider now that the function has more than one variable ( $f(x,y)$ , or  $f(x_1, x_2, \dots, x_n)$ ). We find the gradient of the function  $f$  with respect to  $x$  by varying one variable at a time and keeping the others constant.

For a function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $x \mapsto f(\mathbf{x})$ ,  $\mathbf{x} \in \mathbb{R}^n$  of  $n$  variables  $x_1, \dots, x_n$ , we define the *partial derivatives* as

$$\begin{aligned}\frac{\partial f}{\partial x_1} &= \lim_{h \rightarrow 0} \frac{f(x_1 + h, x_2, \dots, x_n) - f(x)}{h} \\ &\vdots \\ \frac{\partial f}{\partial x_n} &= \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_{n-1}, x_n + h) - f(x)}{h}\end{aligned}$$

and collect them in the row vector

$$\nabla_x f = \text{grad} f = \frac{df}{dx} = \left[ \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right] \in \mathbb{R}^{1 \times n}.$$

The row vector  $\nabla_x f$  is called the *gradient* of  $f$  or the *Jacobian*, and it is defined as the generalization of the derivative to functions of several variables.

## Partial derivatives and gradient

For  $f(x_1, x_2) = x_1^2 x_2 + x_1 x_2^3 \in \mathbb{R}$ , the partial derivatives (i.e., the derivatives of  $f$  with respect to  $x_1$  and  $x_2$ ) are

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = 2x_1 x_2 + x_2^3$$

$$\frac{\partial f(x_1, x_2)}{\partial x_2} = x_1^2 + 3x_1 x_2^2$$

and the gradient is then

$$\frac{df}{dx} = \left[ \frac{\partial f(x_1, x_2)}{\partial x_1} \quad \frac{\partial f(x_1, x_2)}{\partial x_2} \right] = [2x_1 x_2 + x_2^3 \quad x_1^2 + 3x_1 x_2^2] \in \mathbb{R}^{1 \times 2}$$

# Mixed partial derivative

*mixed partials* are defined as the partial derivatives of a partial derivative. For example, for  $f(x,y) = \frac{xy}{z^2} + y\sqrt{z} + \sqrt{xt}$ , the partial derivative with respect to  $z$  is

$$\frac{\partial f}{\partial z} = \frac{-xy}{t^2 z^2} + \frac{y}{2\sqrt{z}}$$

which is still a function of  $x, y, z$ , and  $t$ . Therefore, we can calculate second partial derivatives like so:

$$\frac{\partial^2 f}{\partial x \partial z} = \frac{-y}{t^2 z^2}$$

$$\frac{\partial^2 f}{\partial y \partial z} = \frac{-x}{t^2 z^2} + \frac{1}{2\sqrt{z}}$$

$$\frac{\partial^2 f}{\partial t \partial z} = \left( \frac{-xy}{z^2} \right) \left( -\frac{1}{t^2} \right) = \frac{xy}{t^2 z^2}$$

$$\frac{\partial^2 f}{\partial z^2} = \frac{2xy}{t^2 z^3} - \frac{y}{4z^{3/2}}$$

# Chain Rule for Partial derivatives

To apply the chain rule to partial derivatives, we need to track all the variables. So, if we have  $f(x,y)$ , where both  $x$  and  $y$  are functions of other variables,  $r$  and  $s$ , then we can find the partials of  $f$  with respect to  $r$  and  $s$  by applying the chain rule for each variable,  $x$  and  $y$ . Specifically,

$$\frac{\partial f}{\partial r} = \left( \frac{\partial f}{\partial x} \right) \left( \frac{\partial x}{\partial r} \right) + \left( \frac{\partial f}{\partial y} \right) \left( \frac{\partial y}{\partial r} \right)$$

$$\frac{\partial f}{\partial s} = \left( \frac{\partial f}{\partial x} \right) \left( \frac{\partial x}{\partial s} \right) + \left( \frac{\partial f}{\partial y} \right) \left( \frac{\partial y}{\partial s} \right)$$

Let  $f(x,y) = x^3 + y^3$  with  $x(r,s) = 3r + 2s$  and  $y(r,s) = r^2 - 3s$ . Now find  $\frac{\partial f}{\partial r}$  and  $\frac{\partial f}{\partial s}$ . To find these partials, we'll need to calculate six expressions,

$$\frac{\partial f}{\partial x} = 3x^2, \quad \frac{\partial f}{\partial y} = 3y^2, \quad \frac{\partial x}{\partial r} = 3, \quad \frac{\partial x}{\partial s} = 2, \quad \frac{\partial y}{\partial r} = 2r, \quad \frac{\partial y}{\partial s} = -3$$

## Example

so that the desired partials are

$$\begin{aligned}\frac{\partial f}{\partial r} &= \left(\frac{\partial f}{\partial x}\right) \left(\frac{\partial x}{\partial r}\right) + \left(\frac{\partial f}{\partial y}\right) \left(\frac{\partial y}{\partial r}\right) \\ &= (3x^2)(3) + (3y^2)(2r) = 9x^2 + 6y^2r \\ &= 9(3r+2s)^2 + 6(r^2-3s)^2r = 9(3r+2s)^2 + 6r^3 - 36rs^2 \\ \frac{\partial f}{\partial s} &= \left(\frac{\partial f}{\partial x}\right) \left(\frac{\partial x}{\partial s}\right) + \left(\frac{\partial f}{\partial y}\right) \left(\frac{\partial y}{\partial s}\right) \\ &= (3x^2)(2) + (3y^2)(-3) = 6x^2 - 9y^2 \\ &= 6(3r+2s)^2 - 9(r^2-3s)^2\end{aligned}$$

### Important remark

remember that  $\frac{\partial f}{\partial r}$  tells us how  $f$  will change for a small change in  $r$ . We'll use this fact during gradient descent.

- 1 Linear Algebra
  - Introduction
  - Linear Systems
  - Basic matrix operations (+, -, \*)
  - Determinants and inverses
  - Vector Spaces and Linear Transformations

- 2 Analytic Geometry
  - Normed Vector Spaces
  - Unit vectors and orthonormal basis

- 3 Matrix Decomposition
  - Diagonalisation
  - Analyse en Composantes Principales

- 4 Calculus and Optimization
  - Differentiation
  - Gradient descent

- 5 Probability Theory

- 6 Statistical Modeling

- 7 Statistical Inference

# What is Gradient Descent?

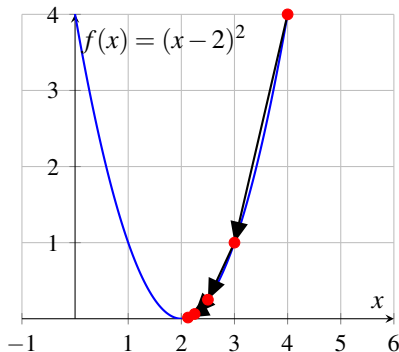
- Gradient Descent is an optimization algorithm used to minimize a function by iteratively moving in the direction of steepest descent as defined by the negative of the gradient.
- Primary use in machine learning: to find the minimum of a loss function.

# One dimensional

We will start by considering gradient descent in one dimension. Assume  $\Theta \in \mathbb{R}$ , and that we know both  $J(\Theta)$  and its first derivative with respect to  $\Theta$ ,  $J'(\Theta)$ . Here is pseudocode for gradient descent on an arbitrary function  $f$ . Along with  $f$  and its gradient  $f'$ , we have to specify the initial value for parameter  $\Theta$ , a step-size parameter  $\eta$ , and an accuracy parameter  $\varepsilon$ :

```
1D-GRADIENT-DESCENT(Theta_init, eta, f, f', epsilon)
\Theta(0) = \Theta_init
t = 0
repeat
    t = t + 1
    Theta(t) = Theta(t-1) - eta * f'(Theta(t-1))
until |f(Theta(t)) - f(Theta(t-1))| < epsilon
return Theta(t)
```

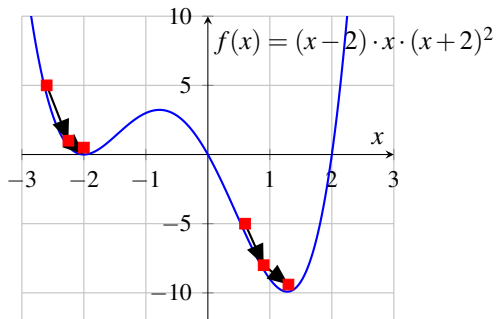
## Gradient Descent on a Convex Function



# Non-Convex

If  $J$  is non-convex, where gradient descent converges to depends on  $\Theta_{\text{init}}$ . When it reaches a value of  $\Theta$  where  $f'(\Theta) = 0$  and  $f''(\Theta) > 0$ , but it is not a minimum of the function, it is called a *local minimum* or *local optimum*.

Minimization of a Non-Convex Function



# Multidimensional

The extension to the case of multi-dimensional  $\Theta$  is straightforward. Let's assume  $\Theta \in \mathbb{R}^m$ , so  $J : \mathbb{R}^m \rightarrow \mathbb{R}$ . The gradient of  $J$  with respect to  $\Theta$  is

$$\nabla_{\Theta} J = \begin{bmatrix} \frac{\partial J}{\partial \Theta_1} \\ \vdots \\ \frac{\partial J}{\partial \Theta_m} \end{bmatrix}.$$

The algorithm remains the same, except that the update step in line 5 becomes

$$\Theta^{(t)} = \Theta^{(t-1)} - \eta \nabla_{\Theta} J(\Theta^{(t-1)})$$

and we have to change the termination criterion. The easiest thing is to keep the test in line 6 as

$$|f(\Theta^{(t)}) - f(\Theta^{(t-1)})| < \varepsilon,$$

which is sensible no matter the dimensionality of  $\Theta$ .

# Mathematical Formulation of Gradient Descent

- Consider a function  $f(\theta)$ , where  $\theta$  represents parameters or weights of the model.
- The gradient (vector of partial derivatives) of  $f$  with respect to  $\theta$  is given by  $\nabla f(\theta)$ .
- Update rule:  $\theta_{\text{new}} = \theta - \eta \nabla f(\theta)$
- Here,  $\eta$  is the learning rate, a scalar that determines the step size.
- **Choosing the Right Learning Rate:**
  - Too small: slow convergence.
  - Too large: may overshoot the minimum.

## Algorithmic Steps

- 1 Initialize parameters  $\theta$  randomly.
- 2 Compute the gradient  $\nabla f(\theta)$ .
- 3 Update the parameters:  $\theta = \theta - \eta \nabla f(\theta)$
- 4 Repeat until convergence or maximum iterations are reached.

# Variants of Gradient Descent

- **Batch Gradient Descent:** Uses the whole dataset to compute the gradient.
- **Stochastic Gradient Descent (SGD):** Uses a single sample per iteration.
- **Mini-batch Gradient Descent:** Uses a subset of samples.

## Remark

Will be discussed in more details in **Machine Learning 1**

# Jacobian

The collection of all first-order partial derivatives of a vector-valued function  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is called the Jacobian. The Jacobian  $J$  is an  $m \times n$  matrix, which we define and arrange as follows:

$$\begin{aligned} J = \nabla f = df(x) &= \left[ \frac{\partial f}{\partial x_1} \cdots \frac{\partial f}{\partial x_n} \right] \\ &= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}, \end{aligned}$$

where

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad J(i,j) = \frac{\partial f_i}{\partial x_j}.$$

The **Jacobian matrix** of a function

$$\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

is a matrix consisting of first-order partial derivatives of the function. If each component of  $\mathbf{f}$  is  $f_i$ , where  $i = 1, \dots, m$ , and if  $\mathbf{f}$  is applied to the vector  $\mathbf{x} \in \mathbb{R}^n$  with components  $x_1, \dots, x_n$ , the Jacobian matrix  $\mathbf{J}$  is defined as:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

# Example

Consider the function  $\mathbf{f}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined by:

$$\mathbf{f}(x_1, x_2) = \begin{bmatrix} x_1^2 x_2 \\ 5x_1 + \sin(x_2) \end{bmatrix}$$

The Jacobian matrix of  $\mathbf{f}$  is given by:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial(x_1^2 x_2)}{\partial x_1} & \frac{\partial(x_1^2 x_2)}{\partial x_2} \\ \frac{\partial(5x_1 + \sin(x_2))}{\partial x_1} & \frac{\partial(5x_1 + \sin(x_2))}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 x_2 & x_1^2 \\ 5 & \cos(x_2) \end{bmatrix}$$

# Applications in Machine Learning

- Machine Learning: to train various types of models (logistic regression, neural networks, etc).
- Optimization: It helps in finding the extremum (maximum or minimum) points of multivariable functions.
- Numerical Analysis: It is used in Newton's method for solving systems of non-linear equations.
- Control Theory: In the analysis and design of control systems, the Jacobian matrix helps study the behavior of complex dynamical systems.



Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong  
Mathematics for Machine Learning (2020)

# The End